

Nrxkqrj Zovmqldoxmev

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ICTS-TIFR

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Decrypting the Title

Nrxkqrj Zovmqldoxmev

Decrypting the Title

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a	b	c	d	e	f	g	h	i	j	k	l
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a	b	c	d	e	f	g	h	i	j	k	l
v	w	x	y	z	a	b	c	d	e	f	g

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a	b	c	d	e	f	g	h	i	j	k	l
v	w	x	y	z	a	b	c	d	e	f	g

Shift: 5

Decrypting the Title

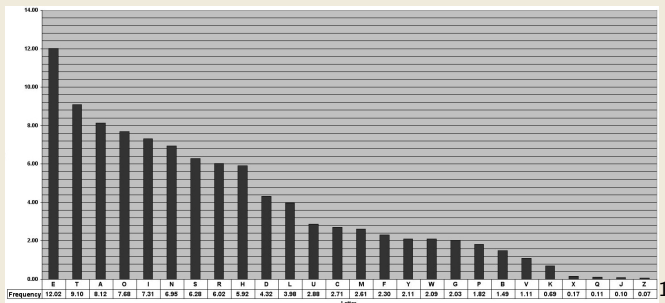
Nrxkqrj Zovmqldoxmev

a	b	c	d	e	f	g	h	i	j	k	l
x	y	z	a	b	c	d	e	f	g	h	i

Shift: 3

Decrypting the Title

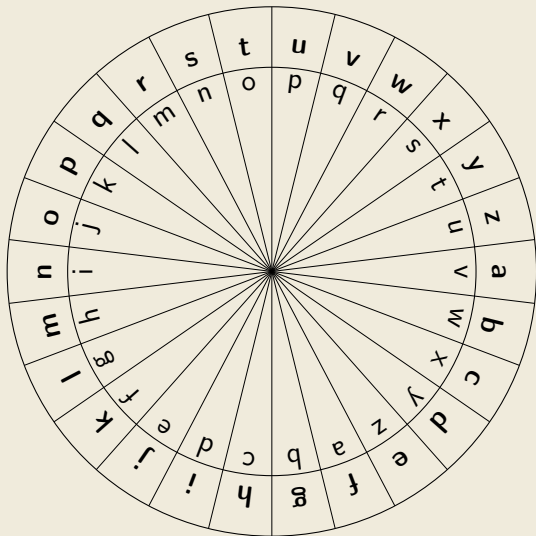
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¹English Letter Frequency (based on a sample of 40,000 words). Source: <https://pi.math.cornell.edu/~mec/>

Decrypting the Title

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Decrypting the Title

- ▶ Caesar cipher (substitution cipher)
- ▶ can be cracked using frequency analysis or brute force computation
- ▶ Key
- ▶ Message

Character encoding

Character encoding

American Standard Code for Information Interchange (ASCII)

UTF-8 (Unicode)

UTF-16

UTF-32

Windows-1252

Character encoding

Convention of using a numerical value to represent each character of a script

American Standard Code for Information Interchange (ASCII)

UTF-8 (Unicode)

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Character encoding

American Standard Code for Information Interchange (ASCII)

A	65	1000001
B	66	1000010
C	67	1000011
⋮	⋮	⋮
a	97	1100001
b	98	1100010
⋮	⋮	⋮

Character encoding

A 65 1000001

B 66 1000010

C 67 1000011

⋮ ⋮ ⋮

a 97 1100001

b 98 1100010

⋮ ⋮ ⋮

$68_{10} :$

2 | 68

2 | 34

2 | 17

2 | 8

2 | 4

2 | 2

2 | 1

0

0

1

0

0

0

1

$\left. \vphantom{\begin{array}{c} 0 \\ 0 \\ 1 \\ 0 \\ 0 \\ 0 \\ 1 \end{array}} \right\} = 1000100_2$

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7. The Receiver uses the key to decrypt the message.

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- ▶ What assumption have we made in this protocol that makes it impractical for real-world communication?

How do they share the key in the first place?

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bit-to-qubit correspondence

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	Horizontal-Vertical basis	$\pm 45^\circ$ basis
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3. The Receiver randomly chooses a basis in which to measure it. Record the basis in the sheet.
4. The Sender prepares the photon in the chosen state and sends it to the Receiver.

State Preparation	$\oplus$ measurement result	$\otimes$ measurement result
$\rightarrow$	$\rightarrow$	Random (R)
$\leftarrow$	$\leftarrow$	R
$\nearrow$	R	$\nearrow$
$\searrow$	R	$\searrow$

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6. The Receiver records the bit obtained from the measurement.

In which cases will the Sender and Receiver share the same bit with 100% certainty?

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4. The Sender prepares the photon in the chosen state and sends it to the Receiver.
5. The Receiver measures the photon using their chosen basis.
6. The Receiver records the bit obtained from the measurement.
7. For each bit, the Sender tells the Receiver which basis was used to prepare the photon, without revealing the bit value.
8. The Receiver compares this with the basis used for the measurement: If the bases match, keep the bit. If the bases differ, discard the bit. The Receiver tells the Sender which bits to keep and which to discard.

What if there is an eavesdropper?

No Cloning Theorem

It is impossible to create an independent and identical copy of an arbitrary unknown quantum state

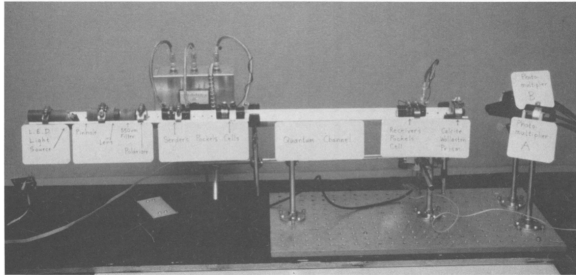


Fig. 2. Photograph of the apparatus.

Incoherent green light flashes are produced by Light Emitting Diode (LED) on the left, collimated into a beam by Pinhole and Lens, then pass through a 550 nm Filter and a horizontal Polarizer. Sender's Pockels Cells convert the horizontal polarization into an arbitrary sequence of the four polarization states (horizontal, vertical, left-circular, and right-circular). After traversing the quantum channel, a 32 cm free air optical path, the beam passes through Receiver's Pockels Cell, which, if energized, converts rectilinear into circular polarizations and vice versa. Finally, a calcite Wollaston prism splits the beam into horizontally and vertically polarized components, in which individual photons are detected by Photomultiplier tubes A and B, respectively.

